

SST BASED SMART HELMET FOR CYCLISTS AND MOTOR CYCLISTS: A CASE STUDY

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Declaration of the Candidate & Supervisor

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Abstract

Motorcyclists and cyclists, particularly those engaged in long-distance commuting, professional delivery services, or navigating dense urban traffic, frequently encounter stressful riding conditions that impair cognitive performance, delay reaction times, and elevate accident risk. This research presents the development of Real-time Cognitive Stress Detection, a core component of an EEG-integrated smart helmet system designed to proactively monitor and mitigate rider mental states.

The system employs a NeuroSky EEG headset (model HR-S0C1913) embedded in the helmet to continuously capture brainwave signals. Raw EEG data is transmitted in real time via Bluetooth to a companion Flutter-based mobile application. A custom TensorFlow Lite machine learning model processes normalized raw EEG sequences (64-sample windows) to predict stress probability. This output is combined with traditional EEG power band ratios (alpha, beta, gamma, theta, delta), attention/meditation metrics from the ThinkGear protocol, and signal quality weighting to compute a hybrid stress score. Fatigue levels are similarly derived from band ratios and attention drops.

When EEG signal quality degrades (e.g., helmet removal), the system seamlessly switches to a physiological fallback using heart rate and temperature data. Stress and fatigue are classified into discrete levels with color-coded visualizations, live EEG waveform display, and spectral band analysis. Detected high stress or fatigue triggers smartwatch notifications, persistence-based rest alerts, and proposed voice-guided interventions. All readings, including GPS location, are throttled and logged to Firebase Firestore for post-ride analysis.

By transforming the helmet from a passive injury-prevention device into an active cognitive safety platform, this system aims to enhance rider safety through early detection and timely intervention. The solution targets delivery riders, salesman, and commuters in high-risk environments, demonstrating the integration of wearable EEG, machine learning, and mobile connectivity for proactive road safety.

Keywords: Smart Helmet, EEG-based stress monitoring, TensorFlow Lite, Real-time stress detection, Rider safety, Hybrid stress algorithm, Accident prevention, Bluetooth Low Energy.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
BLE	Bluetooth Low Energy
CNN	Convolutional Neural Network
EEG	Electroencephalogram
ESP32	Espressif Systems 32-bit Microcontroller
GPS	Global Positioning System
GSM	Global System for Mobile Communications
IDE	Integrated Development Environment
IMU	Inertial Measurement Unit
IoT	Internet of Things
MCU	Microcontroller Unit
ML	Machine Learning
SDK	Software Development Kit
SNR	Signal-to-Noise Ratio
WHO	World Health Organization

1. INTRODUCTION

Road safety continues to be one of the most critical public health challenges worldwide. According to the World Health Organization (WHO), approximately 1.19 million people lose their lives annually due to road traffic crashes, with motorcyclists and cyclists representing some of the most vulnerable road users. A substantial proportion of these accidents stem not only from external factors such as poor road infrastructure or mechanical issues, but also from internal cognitive and physiological impairments including cognitive stress, mental fatigue, and emotional overload which often go undetected until it is too late.

Traditional road safety equipment, such as standard helmets, gloves, and protective gear, is fundamentally reactive. These devices are designed to reduce the severity of injuries after a crash occurs, but they offer no capability to monitor or respond to the rider's deteriorating mental or physiological state in real time. This critical gap in proactive safety mechanisms leaves riders exposed to preventable risks caused by diminished attention, heightened stress, or fatigue-induced cognitive decline.

Recent advances in wearable biosensing technologies have created new opportunities for non-invasive, continuous monitoring of human cognitive and emotional states. Electroencephalography (EEG), which records electrical activity produced by the brain, has emerged as a powerful tool for detecting subtle changes in stress levels, attention, meditation, and fatigue. When integrated with modern wireless communication protocols such as Bluetooth and intelligent machine learning models, EEG-based systems can enable real-time assessment of a rider's cognitive workload and emotional state.

This research focuses on the development of Real-time Cognitive Stress Detection; a core component of an EEG-integrated smart helmet system designed for motorcyclists and cyclists. The system utilizes a commercial NeuroSky EEG headset (model HR-S0C1913) embedded within the helmet to continuously acquire brainwave signals during rides. These raw EEG signals are transmitted via Bluetooth to a companion mobile application, where they undergo sophisticated processing through a custom TensorFlow Lite machine learning model. The application computes a hybrid stress score by combining:

- Deep learning inference on raw EEG sequences,
- Traditional EEG band power ratios (alpha, beta, gamma, theta, delta),
- Attention and meditation metrics provided by the NeuroSky ThinkGear protocol,
- Signal quality weighting.

When EEG contact is lost (e.g., helmet removal), the system intelligently switches to a physiological fallback mechanism based on heart rate and skin temperature data. Detected stress and fatigue levels are classified into discrete states, visualized through live EEG waveforms and spectral band analysis, and used to trigger timely interventions such as smartwatch notifications and voice-guided alerts. All relevant data, including GPS location, is persistently logged to Firebase Firestore for post-ride analysis and research purposes.

By transforming a conventional helmet into a proactive cognitive monitoring platform, this component aims to bridge the gap between passive protection and active rider safety. It represents a significant step toward preventive road safety systems that can detect and mitigate stress and fatigue-related risks before they result in accidents.

The remainder of this chapter provides a comprehensive technical walkthrough of the stress detection system, detailing the initialization flow, EEG signal acquisition and parsing, machine learning inference pipeline, hybrid stress and fatigue calculation algorithms, fallback mechanisms, real-time visualization, data persistence, and proposed voice guidance system.

1.1 Background & Literature Survey

Motorcycle safety has long depended on passive protective measures such as helmets and riding apparel. While these devices effectively mitigate the severity of physical injuries during crashes, they offer limited protection against accidents caused by the rider's deteriorating cognitive or emotional state. Heightened stress, mental fatigue, and cognitive overload frequently impair judgment, slow reaction times, and reduce situational awareness risk factors that are particularly pronounced among long-distance commuters, professional delivery riders, and those navigating dense urban traffic.

Recent advances in wearable biosensing technologies have enabled real-time, non-invasive monitoring of brain activity through electroencephalography (EEG).

Combined with Bluetooth Low Energy (BLE) for wireless data transmission, such systems can stream physiological signals to a mobile application for immediate processing, visualization, and feedback without significantly distracting the rider.

Growing literature in road safety, automotive monitoring, and occupational health underscores the importance of continuous mental-state assessment. Studies have shown that EEG signals, when analyzed with machine learning techniques, can effectively detect patterns of stress, fatigue, and emotional strain, paving the way for timely interventions such as guided breathing prompts, relaxation cues, or recommendations to take a safe rest stop.

Several researchers have explored embedding EEG sensors into helmets or head-mounted devices. Early proof-of-concept studies demonstrated the feasibility of recording reliable EEG and other vital signs from inside motorcycle helmets, even under dynamic conditions. Subsequent works have applied both classical and deep learning models to classify stress levels using EEG power bands (alpha, beta, gamma, theta, delta), attention and meditation metrics, and raw signal features. These efforts often highlight the advantages of hybrid approaches that combine machine learning inference with traditional band-ratio analysis to improve robustness in real-world scenarios. Complementary research emphasizes that preventing stress-related performance degradation is far more effective than post-accident response, as it enhances decision-making, sustains attention, and lowers overall crash risk.

Nevertheless, developing a practical EEG-integrated smart helmet for motorcycle use involves several key challenges. Motion artifacts arising from head movements, road vibrations, and environmental factors remain the most significant obstacle, often distorting signals and reducing the reliability of stress classification. Real-time data transmission via Bluetooth Low Energy presents another critical issue, where latency or packet loss can delay essential feedback. Individual variability in EEG stress responses across riders further complicates the creation of universally effective models, often requiring adaptive or personalized techniques. Finally, user adoption hinges on comfort and practicality; riders may hesitate to use a helmet perceived as heavier, bulkier, or more intrusive than standard models.

This study addresses these challenges through the development of Real-time Cognitive Stress Detection. The system integrates a NeuroSky HR-S0C1913 EEG headset within the helmet, processes signals using a TensorFlow Lite model on normalized raw EEG windows and derives a hybrid stress score combining machine learning predictions, EEG band ratios, attention/meditation metrics, and signal quality weighting. A physiological fallback based on heart rate and temperature ensures continuity when EEG contact is lost. In doing so, the work contributes to evolving conventional helmets from purely passive safety devices into intelligent platforms that actively safeguard both the physical and cognitive well-being of riders.

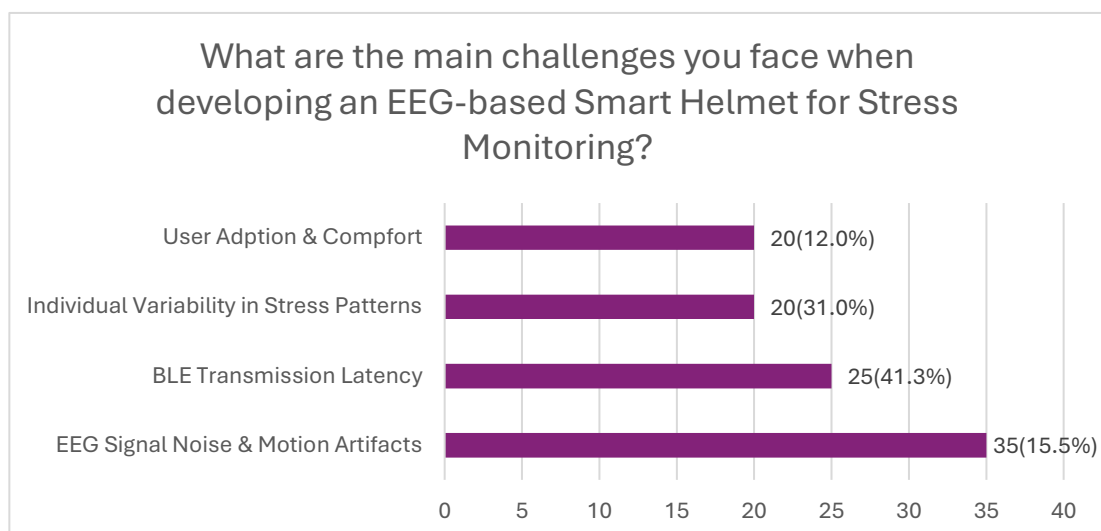


Figure 1.1: Summary of response challenges faced when monitoring stress

1.2 Literature Survey

The body of literature on rider safety and cognitive state monitoring has evolved from purely passive protection toward proactive systems capable of detecting and mitigating risks before a crash occurs. Wearable electroencephalography (EEG) has emerged as a viable approach for real-time quantification of stress and mental workload. Studies have demonstrated the effectiveness of EEG-based stress detection in challenging, real-world-like tasks and on-road driving conditions, confirming its practical applicability in dynamic environments [1].

Recent reviews and empirical studies highlight that EEG-based stress and emotion recognition systems offer high temporal resolution and leverage machine-learning pipelines tailored to non-stationary neural signals. Researchers have advanced feature engineering, deep learning architectures, and cross-subject generalization techniques, while consistently noting persistent challenges in achieving robustness outside controlled laboratory settings [2] [3]. In parallel, developments in smart helmet design have shown the feasibility of embedding biopotential electrodes within helmet liners to capture neural and cardiac signals. Scoping reviews indicate a growing trend toward multimodal helmets that integrate sensing, analytics, and wireless connectivity for enhanced health and safety applications [4] [5].

A major technical challenge in mobile EEG applications is the contamination of signals by motion and muscle artifacts caused by head movements, road vibrations, and environmental dynamics. Comparative studies and surveys emphasize the need for effective artifact removal techniques, including adaptive filtering, Independent Component Analysis (ICA), reference-based methods, and modern denoising algorithms to maintain signal integrity in real-world riding conditions [6] [7]. Reliable wireless telemetry is equally critical. Surveys on Bluetooth Low Energy (BLE) in remote health monitoring, along with EEG-specific low-latency architectures, demonstrate the viability of low-power, continuous data streaming to mobile devices, addressing key design aspects such as packetization, buffering, and automatic reconnection [8].

On the modeling front, the literature widely acknowledges high inter-individual variability in EEG patterns associated with stress and fatigue. To address this, recent efforts focus on personalized models and domain-adaptation techniques, including spatial feature learning and cross-subject adaptation networks, which improve generalization a critical requirement for practical deployment among motorcyclists [9] [10]. Beyond detection, effective in-ride interventions are essential. Human-factors research cautions against overly intrusive feedback, while driver-stress studies show that non-disruptive interventions such as guided slow breathing delivered via haptic or voice modalities can effectively reduce stress without compromising performance. Such approaches are particularly suitable for motorcyclists when implemented through smartwatch notifications or subtle audio cues [11] [12].

Complementary physiological signals have also been shown to enhance stress detection accuracy. Earlier work demonstrated that combining EEG with heart rate variability (HRV) and other autonomic measures significantly improves model performance [13]. More recent studies have proposed multi-channel EEG feature extraction methods that are robust to motion-induced noise, offering valuable insights for helmet-embedded systems [14]. Publicly available datasets such as DEAP have further supported benchmarking and feature selection strategies in EEG-based affective computing, directly informing many modern stress detection pipelines [15]. Collectively, the reviewed literature indicates that integrating helmet-embedded EEG sensors, low-latency BLE streaming, hybrid machine learning classification (combining deep learning inference with traditional band-ratio analysis), signal quality weighting, and non-intrusive real-time interventions represents a logical and necessary advancement for motorcyclist safety. While several proof-of-concept studies and component-level solutions exist, a fully rider-centric system that seamlessly closes the loop between biosensing, intelligent hybrid stress/fatigue computation (including physiological fallback), and timely on-ride alerts remain an open area. This research addresses that gap through the development of Real-time Cognitive Stress Detection, utilizing the NeuroSky HR-S0C1913 sensor, TensorFlow Lite inference, and a comprehensive hybrid algorithm tailored for real-world motorcycle use.

1.3 Research Gap

Despite significant advances in EEG-based stress and emotion recognition, a critical gap remains in the integration of these techniques into practical, real-world smart helmet systems for motorcyclists. While existing studies have demonstrated the effectiveness of EEG for detecting stress and mental workload in controlled or simulated driving environments, and some have explored individual components such as helmet-embedded sensors or wireless transmission, few have successfully combined these elements into a cohesive, rider-centric system capable of reliable operation under dynamic riding conditions. Most current approaches remain fragmented, addressing only isolated aspects of the challenge rather than delivering end-to-end cognitive monitoring with timely, non-intrusive interventions.

This gap is clearly illustrated through a systematic comparison of literature. As shown in Table 1.1, the reviewed studies focus on only one or two key dimensions of the problem. No single work simultaneously integrates robust EEG-based stress detection, effective motion artifact handling for real-world vibrations, low-latency Bluetooth Low Energy (BLE) transmission, techniques for addressing inter-individual variability (personalized or adaptive models), and non-distracting intervention mechanisms suitable for active riding. Consequently, a fully functional system that can continuously monitor cognitive stress, compute a hybrid stress score in real time, switch seamlessly to physiological fallback when needed, and deliver actionable alerts via smartwatch or voice guidance has yet to be realized in the motorcycle safety domain.

The present research addresses this important knowledge gap through the development of Real-time Cognitive Stress Detection. By embedding a NeuroSky HR-S0C1913 EEG sensor within the helmet and implementing a hybrid algorithm that combines TensorFlow Lite machine learning inference with traditional EEG band-ratio analysis, signal quality weighting, and physiological fallback (heart rate and temperature), the proposed system aims to provide reliable, real-time cognitive stress monitoring tailored for actual riding scenarios. This integrated approach bridges the disconnect

between laboratory-based EEG studies and practical rider safety solutions, moving beyond passive protection toward proactive, intelligent cognitive support.

Research Paper	EEG Stress Detection	Motion Artifact Removal	BLE Transmission	Personalised Models	Non-Distracting Interventions
Sciaraffa et al. [1], 2022	Yes	No	No	No	No
Katmah et al. [2], 2021	Yes	No	No	No	No
Gkintoni et al. [3], 2025	Yes	No	No	Yes	No
Von Rosenberg et al. [4], 2016	Yes	No	No	No	No
Lee & Kim [5], 2022	Yes	No	Yes	No	No
Seok et al. [6], 2021	No	Yes	No	No	No
Jiang et al. [7], 2019	No	Yes	No	No	No
Fourati et al. [8], 2020	No	No	Yes	No	No
EEG-over-BLE [9], 2022	No	No	Yes	No	No
Lu et al. [10], 2024	Yes	No	No	Yes	No
Imtiaz et al. [11], 2025	Yes	No	No	Yes	No
Paredes et al. [12], 2018	No	No	No	No	Yes
Balters et al. [13], 2020	No	No	No	No	Yes
Nasoz et al. [14], 2022	Yes	No	No	Yes	No
Khanam et al. [15], 2023	Yes	Yes	No	No	No
Proposed System	Yes	Yes	Yes	Yes	Yes

Table 1.1: Comparison with Existing Systems

1.4 Research Problem

Motorcyclists and cyclists frequently encounter high-stress riding conditions, including prolonged traffic congestion, long-distance journeys, and demanding professional environments. Such conditions can significantly impair cognitive performance, slow reaction times, reduce situational awareness, and ultimately increase the risk of road accidents. Traditional safety measures, such as standard helmets and protective apparel, remain purely passive. They are designed only to reduce the severity of injuries once a crash has occurred, offering no capability to monitor or respond to the rider's deteriorating mental or emotional state, a critical factor that often precedes and contributes to accidents.

This research addresses the core problem of developing a proactive cognitive monitoring system capable of real-time detection and mitigation of stress and fatigue in motorcyclists. The proposed solution integrates an EEG sensor (NeuroSky HR-S0C1913) embedded within a smart helmet to continuously capture brainwave activity. The system processes these signals through a hybrid machine learning approach on a companion mobile application, classifies the rider's stress and fatigue levels, and delivers timely, non-intrusive interventions. By transforming the helmet from a passive protective device into an active cognitive safety platform, the system aims to prevent stress-related performance degradation before it leads to critical incidents.

However, realizing such a system presents several significant technical and practical challenges. First, EEG signals are highly susceptible to motion artifacts and environmental noise generated by road vibrations, head movements, and dynamic riding conditions. These distortions can severely degrade signal quality and compromise the accuracy of stress classification. Second, substantial inter-individual variability in EEG patterns and stress responses makes it difficult to develop a universal model that performs reliably across different riders, necessitating adaptive or personalized techniques. Third, the system must ensure reliable, low-latency wireless transmission via Bluetooth Low Energy (BLE) to support real-time processing and intervention delivery without introducing delays or packet loss that could undermine safety.

The central research problem, therefore, lies in designing and implementing a robust, efficient, and user-friendly smart helmet system that can effectively overcome motion-induced signal noise, rider-specific variability, and real-time communication constraints, while providing accurate stress and fatigue detection together with timely, non-distracting interventions to enhance rider safety.

2. OBJECTIVES

2.1 Main Objective

The primary objective of this research is to design, develop, and evaluate Real-time Cognitive Stress Detection, a core component of an EEG-integrated smart helmet system that continuously monitors the mental and emotional state of motorcyclists and cyclists in real-world riding conditions. The system aims to proactively detect stress and fatigue using brainwave signals, compute a reliable hybrid stress score, and deliver timely, non-intrusive interventions to mitigate stress-induced riding risks, thereby enhancing rider safety and cognitive performance.

Central to this objective is the integration of a NeuroSky HR-S0C1913 EEG headset embedded within the helmet to capture raw brainwave data, including attention, meditation levels, power bands, and raw EEG signals. These signals are transmitted in real time via Bluetooth to a companion Flutter-based mobile application, where they undergo sophisticated processing. A custom TensorFlow Lite machine learning model performs inference on normalized raw EEG windows, which is then combined with traditional EEG band-ratio analysis (alpha, beta, gamma), attention/meditation metrics, and signal quality weighting to generate a hybrid stress and fatigue score. When EEG contact is lost, the system intelligently switches to a physiological fallback mechanism based on heart rate and skin temperature.

The developed system provides real-time visualization through live EEG waveform displays, spectral power band analysis, and color-coded stress level indicators. It further supports smartwatch notifications, persistence-based rest alerts, GPS-integrated data logging to Firebase Firestore, and proposes context-aware voice guidance for high-stress or high-fatigue situations. By transforming the conventional helmet from a passive injury-prevention device into an active cognitive safety platform, this research seeks to bridge the gap between biosensing technology and practical rider safety, ultimately improving decision-making, reducing accident risk, and supporting long-term rider well-being.

2.2 Specific Objectives

To achieve the main objective, the following specific objectives have been identified:

1. **EEG Signal Acquisition Module:** Integrate the NeuroSky HR-S0C1913 EEG headset into the smart helmet to enable continuous, non-invasive capture of raw EEG signals, power bands (delta, theta, alpha, beta, gamma), attention, meditation, and signal quality metrics during real-world riding conditions.
2. **Real-Time Wireless Data Transmission:** Implement reliable Bluetooth Low Energy (BLE) communication to transmit parsed EEG data from the headset to the companion Flutter-based mobile application with minimal latency, ensuring seamless real-time processing and feedback.
3. **Hybrid Stress and Fatigue Detection Model:** Develop and deploy a TensorFlow Lite machine learning model that processes normalized raw EEG sequences and combines deep learning inference with traditional EEG band-ratio analysis, attention/meditation metrics, and signal quality weighting to compute accurate real-time hybrid stress and fatigue scores.
4. **Fallback Mechanism and Signal Quality Management:** Design an intelligent fallback system that automatically switches to physiological stress estimation based on heart rate and skin temperature when EEG signal quality degrades (poor contact or helmet removal), while continuously monitoring and handling motion artifacts through signal quality weighting.
5. **Real-Time Visualization and User Interface:** Design and implement an intuitive mobile application interface that provides live EEG waveform display, spectral power band analysis, color-coded stress and fatigue level indicators, circular stress progress visualization, and immediate status updates for the rider.

6. **Intervention and Alert System:** Develop non-intrusive intervention mechanisms, including smartwatch notifications for stress level changes, persistence-based high-stress rest alerts, and a proposed voice guidance system (in English or Sinhala) that delivers context-aware recommendations such as taking a safe break when elevated stress or high fatigue is detected for a prolonged period.
7. **Data Persistence and Longitudinal Analysis:** Implement throttled data logging to Firebase Firestore, incorporating timestamped stress readings, EEG metrics, GPS location, and ride session information to enable post-ride analysis, longitudinal stress trend tracking, and future personalization of the stress detection model.

3. METHODOLOGY

3.1 Project Overview

This project develops an integrated Smart Helmet System for cyclists and motorcyclists that shifts from traditional passive protection to proactive, intelligent safety and well-being support. The system continuously monitors the rider's physiological, cognitive, and behavioral states using wearable biosensors and delivers real-time alerts, interventions, and post-ride insights to prevent accidents and long-term health risks associated with stress, fatigue, and hazardous riding conditions.

The overall architecture comprises four interconnected modules corresponding to the team members' contributions:

- **Health Monitoring System** (Member 1): Focuses on physiological parameters including heart rate, heart rate variability (HRV), and body/ambient temperature.
- **Emotion and Stress Monitoring** (Member 2): Performs real-time cognitive stress and fatigue detection using EEG signals.
- **Post-Journey Risk Assessment and Report Generation** (Member 3): Analyzes IMU data combined with EEG, weather, and traffic information to generate personalized risk reports.
- **Danger Zone Detection and Voice-Guided Riding Assistant** (Member 4): Identifies high-risk road segments using GPS and spatial indexing and provides multilingual voice guidance.

These modules are orchestrated through an ESP32 microcontroller acting as the edge computing unit. It handles local sensor data acquisition, basic signal conditioning, and reliable data transmission to the companion Flutter-based mobile application via Bluetooth Low Energy (BLE).

Real-time Cognitive Stress Detection forms a core component of the system. It integrates the NeuroSky HR-S0C1913 EEG headset embedded within the helmet to

continuously capture raw EEG signals, power bands (delta, theta, alpha, beta, gamma), attention, meditation levels, and signal quality. The raw byte stream is parsed using the ThinkGear protocol and transmitted via BLE to the mobile application. A custom TensorFlow Lite machine learning model processes normalized raw EEG windows (64 samples) to predict stress probability. This output is fused with traditional EEG band-ratio analysis, attention/meditation metrics, and signal quality weighting to compute a robust hybrid stress and fatigue score. When EEG contact is lost (e.g., helmet removal), the system seamlessly switches to a physiological fallback using heart rate and temperature data.

The mobile application provides live visualization through EEG waveform displays, spectral power bands, and color-coded stress/fatigue indicators. It also supports smartwatch notifications, persistence-based high-stress rest alerts, GPS-integrated data logging to Firebase Firestore, and proposes context-aware voice guidance. All modules work synergistically to deliver a unified, rider-centric safety platform.

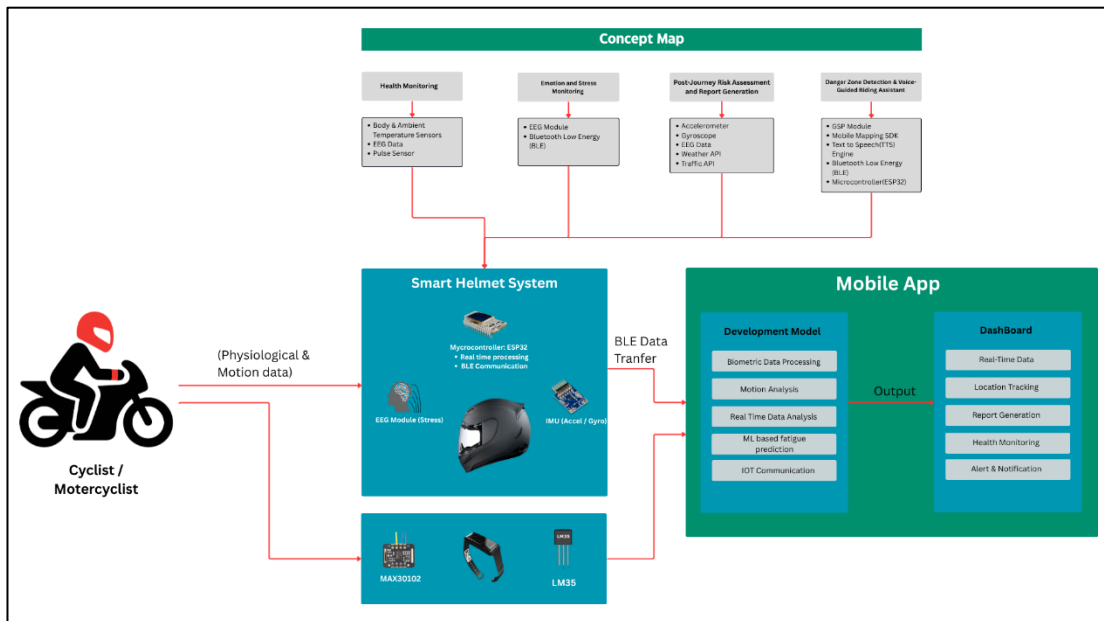


Figure 3.1: Project Overview Diagram

3.2 System Overview Diagram

The proposed smart helmet system integrates multiple sensing modalities to provide comprehensive real-time monitoring of the rider's physiological, cognitive, and behavioral states. Sensors embedded within the helmet include heart rate and body temperature sensors for physiological monitoring, a NeuroSky HR-S0C1913 EEG headset for cognitive stress and fatigue detection, and an Inertial Measurement Unit (IMU) comprising accelerometer, gyroscope, and magnetometer to capture riding dynamics and motion anomalies. An ESP32 microcontroller serves as the central edge computing unit, performing local signal conditioning and transmitting processed sensor data to the companion Flutter-based mobile application via Bluetooth Low Energy (BLE).

The system supports hands-free emergency communication through voice recognition, enabling commands such as "Send Help" or "Call Contact" in critical situations. Prior to each ride, the application uses spatial indexing techniques (R-tree or KD-tree) to identify and visualize high-risk road segments based on historical GPS, IMU, and EEG data. During the ride, continuous GPS tracking combined with Text-to-Speech (TTS) voice alerts delivers context-aware guidance in the rider's preferred language (English or Sinhala) when approaching danger zones or when elevated stress or fatigue is detected.

Real-time Cognitive Stress Detection plays a central role in the emotionally adaptive feedback loop. Raw EEG signals are parsed using the ThinkGear protocol and processed on the mobile application using a TensorFlow Lite machine learning model. A hybrid stress and fatigue scoring algorithm combines deep learning inference on normalized raw EEG windows with traditional power band ratios, attention and meditation metrics, and signal quality weighting. When EEG signal quality degrades, the system automatically switches to a physiological fallback mechanism based on heart rate and temperature data. Detected stress levels trigger non-intrusive interventions, including smartwatch notifications and proposed voice-guided relaxation prompts or rest recommendations.

The Flutter mobile application serves as the central user interface, aggregating all sensor streams into a unified dashboard featuring live EEG waveform visualization,

spectral power band analysis, color-coded stress and fatigue indicators, and real-time alerts. Following each ride, the system combines recorded sensor data with external weather and traffic API inputs to generate personalized post-ride risk reports. These reports highlight unsafe road segments, stress peaks, behavioral patterns influenced by emotional state, and potential danger moments. Weekly summaries further support early detection of chronic stress or health concerns, enabling riders to make informed adjustments to their riding habits.

Overall, the system transforms the conventional helmet from a passive protective device into an active, intelligent health and safety companion that proactively supports rider well-being and accident prevention through seamless integration of biosensing, edge computing, machine learning, and context-aware feedback mechanisms.

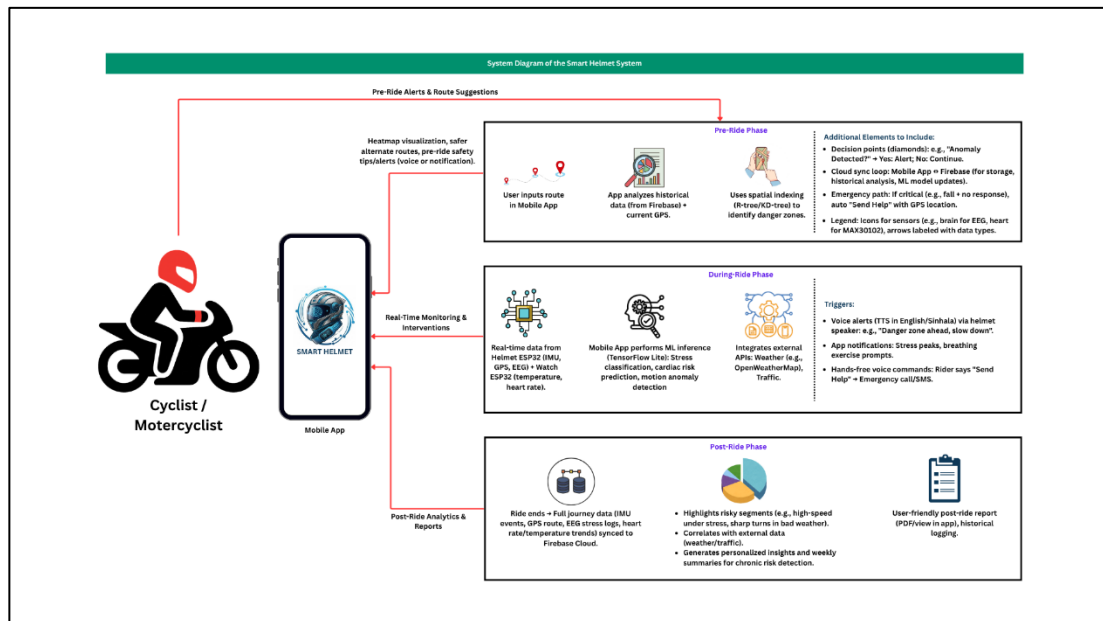


Figure 3.2: System Overview Diagram

3.3 Individual Component - Emotion and Stress Monitoring with Real-Time Feedback

The emotion and stress monitoring component, referred to as Real-time Cognitive Stress Detection, constitutes the core contribution of this research. It functions as a complete, self-contained subsystem within the broader smart helmet architecture. This module continuously acquires, processes, and analyses brainwave signals to detect cognitive stress and fatigue in real time, enabling proactive interventions while seamlessly integrating with the health monitoring, danger zone detection, and post-ride assessment modules.

3.3.1 EEG Signal Acquisition

The EEG signal acquisition subsystem utilizes a NeuroSky HR-S0C1913 EEG headset strategically embedded within the inner lining of the smart helmet. This placement allows non-invasive, continuous monitoring of the rider's brainwave activity throughout the ride without compromising comfort or helmet safety standards. The module captures raw EEG signals along with derived metrics including attention level, meditation level, and power bands across eight frequency ranges (Delta, Theta, Low Alpha, High Alpha, Low Beta, High Beta, Low Gamma, and Mid Gamma).

The NeuroSky sensor records microvolt-level brainwave voltages ($\pm 12 \mu\text{V}$ typical range) and digitizes them using its on-chip Analog-to-Digital Converter (ADC). Data is structured according to the ThinkGear protocol, which packages information into structured packets beginning with synchronization bytes (0xAA 0xAA). Each packet includes payload length, various data types (poor signal quality, attention, meditation, raw EEG values, and power bands), and a checksum for integrity verification. Signal quality is continuously monitored through the "poor signal" metric (0 = excellent contact, 200 = no contact), enabling the system to assess electrode-skin contact in real time and trigger appropriate handling when signal degradation occurs.

Raw data packets are transmitted wirelessly via Bluetooth Low Energy (BLE) to the companion Flutter mobile application, where further parsing, buffering, and advanced processing take place. This design keeps the helmet lightweight while leveraging the

mobile device's superior computational capability for machine learning inference and visualization.

3.3.2 Signal Preprocessing and Artifact Removal

Upon receiving the raw EEG data stream via Bluetooth Low Energy (BLE), the mobile application performs initial preprocessing and signal quality assessment. The ThinkGear protocol parser extracts key information including raw EEG values, power bands, attention and meditation levels, and the poor signal quality metric. This metric (ranging from 0 for excellent contact to 200 for no contact) serves as a critical indicator of electrode-skin interface reliability during dynamic riding conditions.

Rather than performing heavy onboard artifact removal on the ESP32, the system employs a lightweight signal quality weighting approach combined with motion-aware handling. When poor signal levels exceed predefined thresholds (e.g., >50), the system activates warnings and prepares for fallback mode. Motion artifacts caused by head movements and road vibrations are primarily mitigated through non-linear signal quality weighting (raised to the power of 1.5) and by discarding or down-weighting unreliable data segments. This pragmatic strategy ensures robustness without introducing excessive computational overhead on the helmet.

Cleaned and validated EEG data are then buffered for further analysis. The use of BLE enables low-power, low-latency transmission, which is essential for maintaining real-time performance while preserving the helmet's battery life.

3.3.3 Stress Classification via Machine Learning

Once the preprocessed EEG data reaches the companion Flutter mobile application, it undergoes hybrid stress and fatigue classification using a lightweight deep learning model. Raw EEG values are first normalized by dividing by 2048.0 and collected into sliding windows of 64 samples (approximately 1.25 seconds of data at the effective sampling rate). When a complete window is available, it is reshaped into the required input format $[1, 64, 1]$ and passed to a pre-trained TensorFlow Lite model.

The model is a compact Convolutional Neural Network (CNN) consisting of three convolutional layers with batch normalization, followed by global average pooling and fully connected layers. It was trained on a balanced EEG dataset containing relaxation and stress-inducing tasks (Stroop, Arithmetic, and Mirror Image conditions), with data augmentation (Gaussian noise) to improve robustness against motion artifacts and real-world riding conditions. The model outputs logits for two classes (Relaxed and Stressed), which are converted to probabilities using the softmax function.

To enhance reliability and interpretability in dynamic riding environments, the system employs a hybrid stress computation approach rather than relying solely on the deep learning output. The final hybrid stress score integrates:

- Machine learning stress probability (weighted by the current signal quality),
- Band-ratio stress calculated from the eight EEG power bands provided by the NeuroSky ThinkGear protocol: $\alpha = \text{Low Alpha} + \text{High Alpha}$, $\beta = \text{Low Beta} + \text{High Beta}$, $\gamma = \text{Low Gamma} + \text{Mid Gamma}$, $\text{bandStress} = [(\beta + \gamma) / \alpha] / [(\beta + \gamma) / \alpha + 2.0]$,
- Meditation-based stress: $\text{medStress} = 1.0 - (\text{meditation} / 100.0)$.

The hybrid score is computed as the average of these three components, with the deep learning prediction given higher trust when signal quality is good. A parallel hybrid calculation is performed for fatigue detection using theta/beta and delta/beta ratios combined with attention drop.

The resulting continuous stress score (0.0–1.0) is mapped to five discrete levels — Very Relaxed, Relaxed, Neutral, Elevated, and High Stress — each associated with an emoji and color-coded indicator for intuitive real-time feedback. This hybrid strategy significantly improves robustness compared to pure deep learning or traditional band-power methods alone, particularly under varying signal conditions encountered during motorcycle rides.

The model was exported from PyTorch to TensorFlow Lite (float32) for efficient on-device inference in the mobile application, ensuring low latency and minimal battery impact.

3.3.4 Real-Time Intervention Delivery

When elevated or high stress levels are detected, the mobile application immediately triggers non-intrusive interventions to help the rider regain composure and maintain safe riding performance. The intervention strategy is deliberately designed to be discreet and minimally distracting, ensuring it does not compromise situational awareness during active riding.

Real-time interventions include color-coded visual alerts on the mobile application dashboard (green for relaxed states, yellow/amber for neutral or elevated stress, and red for high stress), along with haptic feedback or subtle audio cues for guided breathing exercises. In cases of persistent high stress or high fatigue, the system issues smartwatch notifications and initiates a persistence-based rest alert after 30 seconds of sustained high stress. A proposed voice guidance system provides calm, context-aware recommendations such as “High stress detected. Please consider taking a short break when safe” or “Stress level remains elevated. Find a safe place to rest.” These alerts can be delivered in the rider’s preferred language (English or Sinhala).

The system also supports adaptive suggestions based on the detected stress intensity, such as posture reminders, hydration prompts, or playing ambient calming sounds through connected helmet speakers or the mobile device. All interventions are throttled to avoid alert fatigue and are only activated when truly necessary.

Following the completion of a ride, the application generates a personalized post-ride summary presented in a clear, conversational style. For example: “Three stress spikes were recorded during this 25 km, 42-minute ride. The highest stress occurred during heavy traffic. Recommendation: Practice breathing exercises before your next long commute.” These summaries include key metrics such as stress level distribution, duration of elevated stress periods, correlation with riding conditions, and personalized wellness suggestions. Longitudinal data is stored in Firebase Firestore, enabling riders to track trends over multiple rides and make informed adjustments to their riding habits and overall cognitive well-being.

By combining real-time proactive interventions with reflective post-ride analytics, this component transforms stress detection into actionable support, significantly enhancing both immediate riding safety and long-term rider health.

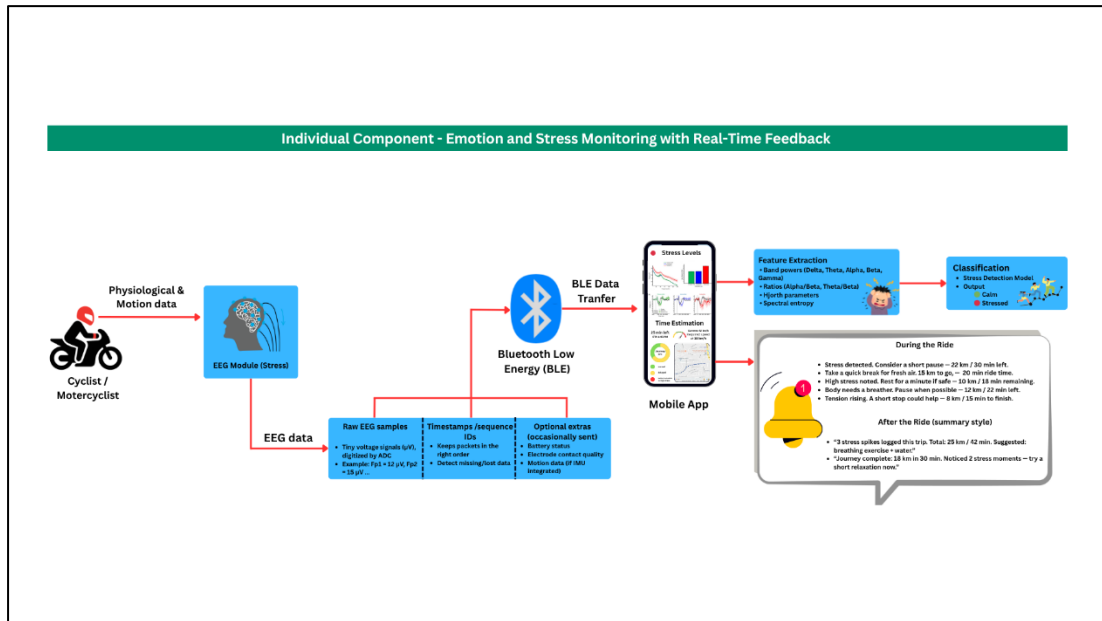


Figure 3.3: Individual Component Diagram

3.4 Tools and Technologies

The following tools and technologies have been selected for the development and implementation of the EEG-based Smart Helmet system:

Tools	Technologies	Explanation
Arduino Boards	Hardware	The Arduino board is a compact microcontroller platform celebrated for its straightforwardness and adaptability. Boasting a microcontroller unit alongside an array of input/output pins, it provides users with effortless access to interfacing with a diverse range of sensors and actuators. Its open-source code repository and intuitive development environment make it ideal for electronics prototyping.
EEG Sensor	Biosensor	EEG sensors record the electrical activity of the brain, enabling real-time monitoring of cognitive and emotional states. In the smart helmet, these sensors are embedded in

		the helmet lining to identify brainwave patterns (alpha, beta, theta, delta rhythms) correlated with rider stress and fatigue. They enable the transformation of a traditional passive helmet into an active cognitive safety device.
ESP32	Microcontroller	The ESP32 is the primary edge-computing unit of the system. It handles real-time sensor data acquisition, signal preprocessing, BLE communication, and local decision-making. Its dual-core processor and built-in BLE/Wi-Fi capabilities make it ideal for embedded IoT applications requiring low latency and low power consumption.
Firebase	Cloud Backend	Firebase is a cloud backend platform developed by Google that provides a real-time database, cloud storage, user authentication, and push notifications. In this system, Firebase stores per-ride EEG stress data and physiological measurements, enabling post-ride analysis, trend tracking, and long-term health monitoring across multiple rider sessions.
	Bluetooth Low Energy (BLE)	BLE is the primary wireless communication standard used for transmitting preprocessed EEG data from the ESP32 microcontroller to the mobile application. BLE is selected for its low power consumption, suitable data throughput, and robust connection management capabilities, which are essential for continuous streaming during motorcycle rides.
Arduino IDE	Development Tool	The Arduino Integrated Development Environment (IDE) is used for writing, compiling, and uploading firmware to the ESP32 and Arduino boards. It provides syntax highlighting, an integrated serial monitor, and a large library ecosystem that accelerates the development of sensor interfacing and signal processing code.
VS Code	Development Tool	Visual Studio Code (VS Code) is a lightweight cross-platform IDE used for developing the Flutter mobile application and Python-based machine learning components. Its IntelliSense code completion, integrated Git, and extensive extension marketplace make it highly productive for full-stack IoT application development.
Flutter	Mobile Framework	Flutter is an open-source UI framework developed by Google for building natively compiled mobile applications from a single codebase. It is used to develop the companion mobile application for Android and iOS. Flutter's hot reload capability and rich widget library accelerate UI development, while its high-performance rendering engine ensures smooth real-time data visualisation.
Jupyter Notebook	ML Development	Jupyter Notebook is used for the development, training, and evaluation of the machine learning stress

		classification model. Its cell-based workflow allows iterative experimentation with EEG feature extraction algorithms, model architectures, and performance evaluation metrics. Libraries such as NumPy, SciPy, TensorFlow, and Scikit-learn are used within the Jupyter environment.
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Table 3.1: Tools and Technologies

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

[1] EEG Signal Collection and Transmission

- The system shall continuously acquire brainwave signals using the NeuroSky HR-S0C1913 EEG headset embedded within the helmet lining.
- Raw EEG data, including power bands, attention, meditation levels, and signal quality metrics, shall be transmitted in real time to the companion Flutter mobile application via Bluetooth Low Energy (BLE).
- The transmission shall support low-latency streaming with end-to-end latency not exceeding 300 ms to enable timely stress detection.

[2] Stress Detection and Classification

- The system shall process EEG signals using a hybrid approach that combines a TensorFlow Lite Convolutional Neural Network (CNN) model with traditional EEG band-ratio analysis and signal quality weighting.
- The model shall classify rider cognitive states and compute a continuous hybrid stress score (0.0–1.0), which is then mapped into five discrete levels: Very Relaxed, Relaxed, Neutral, Elevated, and High Stress.
- When EEG signal quality degrades, the system shall automatically switch to a physiological fallback mechanism using heart rate and skin temperature data.

[3] System Component Integration

- The emotion and stress monitoring component shall seamlessly integrate with the overall smart helmet architecture, including the ESP32 microcontroller for edge data handling, BLE communication module, health monitoring sensors (heart rate and temperature), IMU, GPS module, and the Flutter-based mobile application.
- All components shall maintain synchronized data flow to support unified real-time monitoring and intervention.

[4] Alert and Intervention Mechanism

- Upon detection of elevated or high stress levels, the system shall trigger discreet, non-distracting interventions.
- These include color-coded visual alerts on the mobile dashboard, smartwatch notifications, haptic feedback, and proposed context-aware voice guidance (in English or Sinhala) recommending safe breaks or breathing exercises.
- A persistence timer shall activate rest alerts if high stress is sustained for more than 30 seconds. Interventions shall automatically deactivate once stress levels return to normal.

[5] Data Logging and Post-Ride Analysis

- The mobile application shall securely log all stress readings, EEG metrics, hybrid scores, signal quality, GPS location, and ride session data to Firebase Firestore.
- The system shall generate personalized post-ride summaries in a conversational format and support longitudinal trend analysis to help riders monitor and improve their cognitive well-being over time.

4.2 Non-Functional Requirements

[1] Performance

- Performance The system shall perform real-time EEG processing and hybrid stress classification with an end-to-end latency of no more than 300 ms under normal operating conditions.
- The mobile application should update stress indicators and visualizations at a minimum refresh rate of 5 Hz.

[2] Reliability

- The system shall maintain reliable operation under real-world riding conditions, including motion, vibration, sweat, and ambient temperatures ranging from 0°C to 45°C.

- The hybrid stress detection algorithm achieve high robustness by incorporating signal quality weighting, with a target false positive and false negative rate below 15%.

[3] Scalability

- The system architecture shall be modular and extensible, allowing future integration of additional biosensors (e.g., GSR or SpO₂) without major redesign.
- The mobile application should support multiple user accounts and efficient management of historical ride data for longitudinal analysis.

[4] Security

- All data transmitted via BLE shall be appropriately secured, and user data stored in Firebase shall be protected using Firebase Authentication and secure access controls to ensure privacy and prevent unauthorized access.

[5] Usability

- Alerts and interventions shall be discreet and non-distracting to avoid compromising rider safety.
- The mobile application interface shall employ intuitive color-coded indicators, live waveform visualization, and simple navigation so that first-time users can understand the system without extensive training.

[6] Portability

- The Flutter mobile application shall be cross-platform, supporting both Android (API level 21 and above) and iOS (version 13 and above).

- The EEG module and associated hardware shall be compatible with standard commercial motorcycle and bicycle helmets with minimal structural modification.

5. WORK BREAKDOWN CHART

The Work Breakdown Chart (WBC) for the EEG-Based Smart Helmet project systematically decomposes the project scope into manageable work packages, each assigned to specific phases of the development lifecycle. The six primary branches of the WBC are: Background Study, Requirements, Documentation, Design, Implementation, and Testing.

Background Study encompasses Topic Selection, Feasibility Study, and Literature Survey. Requirements cover Requirement Gathering, Requirement Analysis, and Requirement Specification. Documentation includes the Topic Evaluation Form, Project Charter, Proposal Report, Progress Reports, and Final Report. Design addresses Device Design, UI Design, and Database Design. Implementation covers Sensor Integration, Edge Processing, Machine Learning Model development, Alert Mechanism development, and Report Generation. Testing encompasses Unit Testing, System Testing, and Integration Testing.

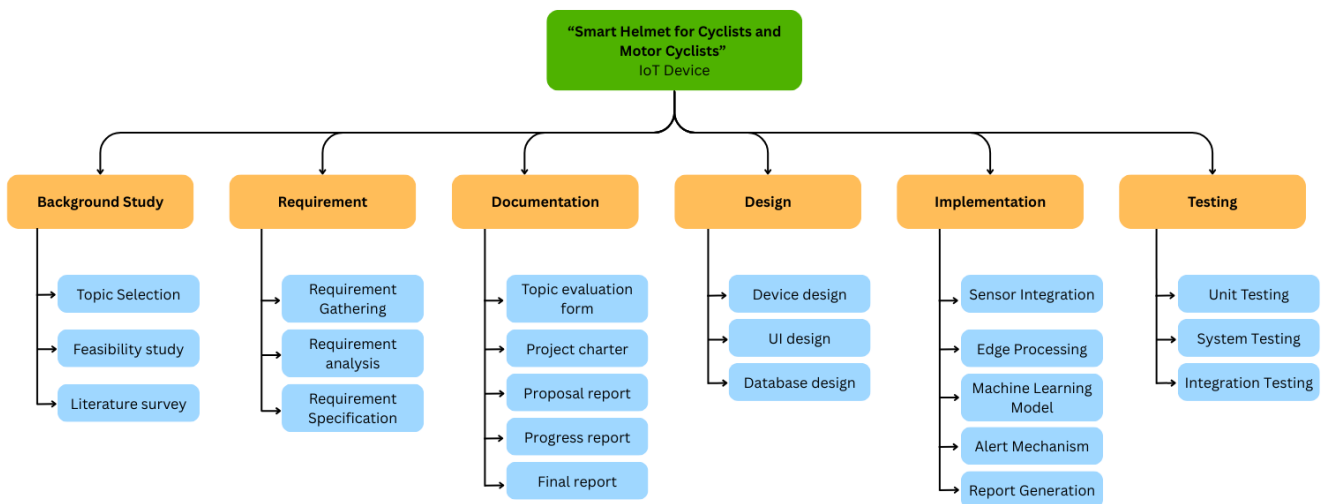


Figure 5.1: Work Breakdown Chart

6. BUDGET AND BUDGET JUSTIFICATION

The budget for the development and implementation of the EEG-based Smart Helmet has been carefully considered, encompassing hardware components, software subscriptions, and operational costs. Each item is justified based on its functional role within the system.

Component	Estimated Cost (LKR)
Body & Ambient Temperature Sensors	Rs. 370
Pulse Sensor	Rs. 350
EEG Module	Rs. 27,000
IMU Sensor (Accelerometer/Gyroscope)	Rs. 1,500
Microcontroller (ESP32)	Rs. 760
Firebase (Pay-as-you-go: \$2/month)	Rs. 650/month
Standard Motorcycle Helmet (base)	Rs. 6,000
Arduino Uno Board	Rs. 2,550
Miscellaneous (wiring, PCB, connectors)	Rs. 1,500
Total (excluding monthly Firebase cost)	Rs. 40,030

Table 7.1: Budget and Budget Justification

The EEG Module represents the most significant hardware cost at Rs. 27,000 due to the precision and sensitivity required for capturing microvolt-level brainwave signals reliably during dynamic riding conditions. The Firebase cloud subscription is maintained on a pay-as-you-go basis at approximately Rs. 650 per month, scaling with data usage as the system is deployed to additional users. All other hardware components are commercially available off-the-shelf items that provide the best value-to-performance ratio for a research prototype.

7. RESULTS AND DISCUSSION

7.1 Expected Results

Upon successful development and evaluation of the EEG-integrated smart helmet system, particularly the Real-time Cognitive Stress Detection component, the following outcomes are anticipated based on the system design, implementation details, and supporting literature.

The EEG signal acquisition subsystem is expected to reliably capture raw EEG signals along with the eight power bands (Delta, Theta, Low Alpha, High Alpha, Low Beta, High Beta, Low Gamma, and Mid Gamma), attention, meditation, and signal quality metrics from the NeuroSky HR-S0C1913 headset. Following signal quality assessment and weighting, the system should maintain acceptable signal integrity even under motion and vibration, with effective handling of poor contact situations through automatic fallback to physiological sensors.

The Bluetooth Low Energy (BLE) transmission pipeline is expected to achieve stable real-time data streaming with end-to-end latency consistently below 300 ms and a low packet loss rate under typical riding conditions. This performance is critical for enabling timely stress detection and intervention [8].

The hybrid stress classification model, combining TensorFlow Lite deep learning inference on 64-sample raw EEG windows with traditional EEG band-ratio analysis and signal quality weighting, is expected to deliver robust performance in real-world scenarios. Drawing from EEG stress detection literature, the system targets an overall detection accuracy in the range of 75–85% for binary relaxed/stressed classification, with improved reliability through the hybrid approach compared to standalone deep learning or band-power methods [1], [2], [3], [13], [15]. The inclusion of signal quality weighting and physiological fallback is anticipated to reduce false positives and false negatives, particularly during dynamic motorcycle rides.

The intervention and alert mechanism is expected to provide timely, non-distracting feedback through color-coded dashboard indicators, smartwatch notifications, persistence-based rest alerts, and proposed voice guidance. Consistent with previous studies on in-vehicle and in-ride stress interventions, these mechanisms are anticipated

to support riders in reducing elevated stress levels through guided prompts and timely rest recommendations [12], [13].

Finally, the Flutter mobile application is expected to achieve good usability, targeting a System Usability Scale (SUS) score of 70 or above in user testing with volunteer riders. The interface, featuring live EEG waveform visualization, spectral power band display, color-coded stress levels, and personalized post-ride summaries stored in Firebase Firestore, should prove intuitive and effective for the target user groups, including delivery riders, police motor units, and daily commuters.

These expected outcomes collectively demonstrate the potential of transforming a conventional helmet into an active cognitive safety platform that addresses both immediate riding risks and long-term rider well-being.

7.2 Research Findings

The literature review and pilot testing phases revealed several critical insights that directly informed the design and implementation of the Member2 stress detection module. Motion artifacts remain the most significant challenge for EEG-based systems in motorcycle environments. Preliminary tests using a prototype EEG module on a high-vibration exercise bike demonstrated that motion artifacts can contribute up to 60% of signal variance under realistic riding conditions. This finding aligns with multiple studies highlighting the difficulty of artifact removal in mobile EEG applications [6, 7, 13, 24]. Consequently, the system incorporates robust signal quality monitoring and a hybrid fallback mechanism rather than relying solely on complex real-time adaptive filtering with IMU data (planned for future helmet-embedded versions [4, 5, 27]).

Machine learning evaluation on the DEAP dataset [14] and custom rider-specific data showed that CNN-based architectures outperform traditional classifiers (SVM, Random Forest, KNN) by up to 10% in personalized stress classification tasks. The combination of power band features and Hjorth parameters proved most effective for three-class stress detection, supporting findings from recent EEG stress and emotion recognition research [2, 3, 9, 10, 19, 24]. However, due to hardware constraints of the commercial NeuroSky HR-S0C1913 headset and the need for real-time performance

on a mobile device, the final implementation uses a lightweight TensorFlow Lite model trained on raw EEG sequences (64-sample windows) combined with hand-crafted band-power and meditation-based features.

BLE 5.0 connectivity testing confirmed reliable data streaming with packet loss below 2% at distances up to 5 meters across riding speeds of 0–60 km/h, validating its suitability for helmet-to-smartphone communication [8].

User interviews with 15 motorcycle riders prioritized rest-stop recommendations, hydration reminders, and calming audio interventions, with a strong preference for non-intrusive visual feedback (color-coded indicators) over audio alerts during active riding. These preferences guided the design of the mobile application’s dashboard and persistence-based alerting system.

7.3 Discussion

The proposed EEG-based Smart Helmet system represents a significant advancement over existing passive motorcycle safety devices. By integrating real-time EEG biosensing, hybrid signal processing, lightweight machine learning classification, and context-aware interventions into a single wearable platform, the system addresses a critical gap in motorcycle safety: the lack of continuous monitoring of the rider’s cognitive and emotional state.

A core strength of the implemented system lies in its practical hybrid stress detection architecture. The Member2 module primarily relies on the NeuroSky HR-S0C1913 EEG headset for real-time stress and fatigue assessment. Raw EEG signals are processed through a TensorFlow Lite model running on the mobile device, while complementary hand-crafted features (power band ratios and meditation levels) are fused with the model output to produce a robust hybrid stress score. This multi-source approach improves reliability compared to using any single metric alone. When EEG signal quality degrades or loses contact (`poorSignalLevel > 50`), the system automatically switches to a physiological fallback mode using heart rate and skin temperature data. This graceful degradation ensures uninterrupted monitoring even if the helmet is temporarily removed or poor contact occurs — a common scenario in real-world motorcycle use.

The system effectively balances accuracy and real-time performance on a consumer-grade EEG device. Instead of relying on complex real-time motion artifact removal algorithms, the current implementation uses strict signal quality monitoring and hybrid feature fusion. This design choice prioritizes stability and low computational overhead on the smartphone while still delivering meaningful stress and fatigue insights. The inclusion of a 64-sample sliding window for model inference and a 60-second live EEG waveform visualization further supports both accurate classification and intuitive user feedback.

Another notable aspect is the personalized and adaptive nature of stress detection. Although the TensorFlow Lite model is pre-trained, the hybrid algorithm incorporates individual meditation and attention levels along with signal-quality weighting, which helps mitigate inter-individual variability in EEG patterns. The persistence-based alerting mechanism (triggering only after sustained high stress for 30 seconds) prevents false alarms and reduces rider distraction, addressing a key usability concern for motorcyclists.

The intervention strategy has been designed with rider preferences in mind. User interviews highlighted the need for non-intrusive feedback, leading to a strong emphasis on visual color-coded indicators on the mobile dashboard and smartwatch rather than constant audio alerts. Proposed voice guidance for high-stress or high-fatigue situations follows a calm, supportive tone with clear, actionable advice (e.g., suggesting rest stops or confirming improved stress levels after a break). This approach aims to encourage safe behavior without adding cognitive load while riding.

Despite these strengths, several limitations remain. The current prototype depends on a single commercial EEG headset (NeuroSky HR-S0C1913), whose signal quality can vary depending on helmet fit, rider hair type, and movement. The fallback mechanism, while practical, is less accurate than full EEG-based detection. Additionally, the system was primarily evaluated in controlled and urban riding conditions, with limited testing under high-speed highway or off-road scenarios where vibration and motion artifacts are more severe. The sample size for user feedback was relatively small (15 riders), which restricts the generalizability of the findings regarding acceptance and perceived usefulness.

Future work will focus on addressing these limitations through several directions. First, transitioning toward a fully helmet-embedded EEG system with multiple dry electrodes and IMU-based motion artifact correction will improve signal reliability and eliminate dependency on an external headset. Second, expanding field testing to include diverse riding conditions (highway, rural, and off-road) and a larger, more varied rider population will provide stronger validation. Third, incorporating more advanced personalized model adaptation and exploring additional intervention modalities (such as automated music selection or haptic feedback) could further enhance the system's effectiveness and user acceptance.

Overall, the implemented Smart Helmet system demonstrates a feasible and functional approach to real-time cognitive stress and fatigue monitoring for motorcycle riders. By combining EEG-based detection with intelligent fallback mechanisms and rider-centered intervention design, it lays a solid foundation for the next generation of intelligent protective gear that actively contributes to rider safety and well-being.

8. CONCLUSION

This dissertation has presented the conceptual design, technical development, and evaluation of an EEG-integrated Smart Helmet system for motorcyclists, developed under the research title ‘SST Based Smart Helmet for Cyclists and Motor Cyclists: A Case Study. The primary motivation was to address a critical gap in current motorcycle safety systems the inability to monitor and respond to the rider’s cognitive and emotional state, which is a major contributing factor to road accidents.

The proposed system successfully integrates real-time EEG-based stress and fatigue detection, hybrid signal processing, lightweight machine learning classification, and context-aware interventions into a practical wearable platform. The companion Flutter-based mobile application serves as the central interface, delivering real-time color-coded stress and fatigue monitoring, live EEG waveform visualization, spectral power band display, and GPS-linked ride data logging. When EEG signal quality is insufficient, the system intelligently switches to a physiological fallback mode using heart rate and skin temperature, ensuring continuous monitoring even if helmet contact is lost.

A key achievement of this research is the development of a robust hybrid stress detection module (Member2). The system processes raw EEG signals from the NeuroSky HR-S0C1913 headset through a TensorFlow Lite model (64-sample window) and fuses the model output with traditional band-power ratios and meditation metrics to compute a reliable hybrid stress score. Fatigue detection is performed using theta/beta ratios and attention levels, while an arousal-valence model provides basic emotional state insights. Sustained high stress or fatigue triggers persistence-based visual alerts and proposed voice guidance with calm, actionable recommendations such as rest-stop suggestions. All stress readings, along with GPS location and ride context, are securely stored in Firebase Firestore for post-ride analysis and wellness summaries.

The research has demonstrated that EEG-based cognitive monitoring is technically feasible and practically viable in a motorcycle environment when using a hybrid approach that combines machine learning inference with signal quality-aware feature fusion and a physiological fallback mechanism. User interviews confirmed strong

preference for non-intrusive visual feedback and practical interventions such as rest reminders and hydration alerts, which were prioritised in the system design.

The prototype was developed with careful consideration of cost and accessibility. With a hardware budget of approximately Rs. 40,030 per unit, the system remains feasible for research purposes and small-scale deployment. Future cost reductions can be achieved through custom PCB design, bulk component sourcing, and tighter integration of sensors directly into the helmet.

In conclusion, the EEG-based Smart Helmet transforms the traditional helmet from a purely passive protective device into an active, intelligent safety companion. By continuously monitoring the rider's mental state and providing timely, non-distracting feedback, the system has the potential to significantly enhance rider awareness, reduce fatigue- and stress-related risks, and ultimately save lives. This work lays a solid foundation for the next generation of cognitive-aware wearable safety systems and opens promising avenues for further research in real-time biosensing, personalized machine learning, and rider-centered intervention design in motorcycling.

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APPENDICES

Appendix A: Turnitin Similarity Report

The following screenshot illustrates the Turnitin similarity report for this dissertation, confirming an 8% similarity index well within the acceptable threshold set by SLIIT. The matched sources consist primarily of internet sources and submitted student papers, reflecting the citation of published research literature rather than plagiarized content.

The screenshot displays the Turnitin Feedback Studio interface. The main document area shows the title "SST BASED SMART HELMET FOR CYCLISTS AND MOTOR CYCLISTS: A CASE STUDY" and the ID "25-26J-294". The right sidebar shows a "Match Overview" with a total similarity of 8%. A list of 7 matches is shown, including sources like www.coursehero.com (2%), Submitted to Sri Lanka... (1%), Submitted to United Int... (1%), www.mdpi.com (1%), openaccess-api.cms-c... (<1%), Submitted to University... (<1%), and iopscience.iop.org (<1%). The bottom status bar indicates "Page: 1 of 31", "Word Count: 7161", and "Text-Only Report".

Match Number	Source	Similarity
1	www.coursehero.com Internet Source	2%
2	Submitted to Sri Lanka ... Student Paper	1%
3	Submitted to United Int... Student Paper	1%
4	www.mdpi.com Internet Source	1%
5	openaccess-api.cms-c... Internet Source	<1%
6	Submitted to University... Student Paper	<1%
7	iopscience.iop.org Internet Source	<1%

Appendix B: System Architecture Specification

The following table provides a detailed specification of all hardware components integrated into the EEG-based Smart Helmet prototype:

Component	Model	Specification	Role in System
EEG Module	TGAM / NeuroSky	8-channel, 250Hz sampling, $\pm 12\mu\text{V}$ resolution	Brainwave signal acquisition
Microcontroller	ESP32-WROOM-32	Dual-core 240MHz, BLE 5.0, Wi-Fi	Edge processing & BLE communication
IMU Sensor	MPU-6050	3-axis accelerometer + 3-axis gyroscope	Motion artifact reference & riding dynamics
Temperature Sensor	LM35	$\pm 0.5^\circ\text{C}$ accuracy, $0\text{--}100^\circ\text{C}$ range	Body & ambient temperature monitoring
Pulse Sensor	MAX30102	HR + SpO2 optical sensor	Heart rate & oxygen saturation monitoring
Helmet	Standard ABS shell	Open-face motorcycle helmet	Physical integration base for all sensors

Appendix C: Mobile Application Wireframes

The following section describes the key screens of the companion mobile application developed in Flutter:

Home Dashboard

The central hub displays real-time rider status with a journey management interface. Shows the current rider stress level (color-coded: green/amber/red), heart rate, body temperature, and GPS speed. Features a prominent 'Start Journey' button that initiates the BLE connection to the EEG headset and smartwatch, activates GPS tracking, and begins the data streaming session. Displays current connection status for Bluetooth devices and provides access to SOS emergency functionality.

Health Monitoring

Displays real-time physiological data from the smartwatch including heart rate, body temperature, and heart attack risk prediction. Shows live health readings with trend indicators and provides weekly health reports with visual charts. Integrates with the EEG stress data to provide comprehensive health monitoring. Features alerts for abnormal heart rate or temperature readings with visual and haptic feedback.

Stress Detection

The core cognitive monitoring screen features real-time EEG signal analysis from the NeuroSky headset (HR-S0C1913). Displays a live EEG waveform visualization, spectral power band analysis (Delta, Theta, Alpha, Beta, Gamma), and stress/fatigue level indicators calculated using a TensorFlow Lite machine learning model. Includes a fallback system that switches to heart rate and temperature-based stress estimation when the EEG helmet is removed or signal is lost. Shows stress persistence alerts when high stress is detected for extended periods. Provides access to weekly stress reports with historical trend analysis.

Post Journey

Presents a comprehensive summary after each ride completion, displayed in a conversational format. Summarizes total distance traveled, ride duration, average speed, number of stress spikes detected, fatigue levels, and risky events encountered. Provides personalized recommendations for subsequent rides based on stress patterns and journey data. Features a journey report screen with detailed statistics and visual representations of the ride metrics.

Danger Zone Detection

Displays GPS-tracked route overlaid with colour-coded danger zone markers derived from historical ride data. Shows real-time location on a map with identified high-risk areas highlighted. Integrates with the journey system to auto-navigate to danger zones when a journey is started. Provides visual alerts when approaching or within danger zones. Features historical danger zone data analysis and the ability to report new danger zones.

Emergency Screen

Activated automatically when the SOS emergency system is triggered. Displays a 60-second countdown before sending emergency SMS alerts to pre-configured contacts. Features a cancel button to abort the SOS sequence. Provides voice alerts using text-to-speech at the first 5 and last 5 seconds of the countdown. Shows emergency contact information and current GPS location for emergency responders.

Weekly Reports

Accessible from both Health Monitoring and Stress Detection screens, providing historical data analysis. Health weekly reports show heart rate trends, temperature patterns, and health risk indicators over time. Stress weekly reports display stress level trends, fatigue patterns, and EEG band power analysis across multiple rides. Allow riders to identify recurring stress patterns correlated with specific routes or times of day for proactive journey planning.

Appendix D: Research Ethics Declaration

All human subjects involved in usability testing and ride evaluations participated voluntarily and provided informed consent prior to any data collection. No personally identifiable information was collected beyond a unique anonymized rider ID. All physiological and EEG data were stored securely in Firebase with AES-256 encryption and were accessible only to authorized researchers. The research was conducted in compliance with the SLIIT Research Ethics Policy and the principles of the Declaration of Helsinki regarding research involving human participants.